

Customer Support Enhancement in E-Commerce with GenAI and Recommendation Systems: Evaluating Large Language Models

Project Advisor: Dr. Magdalini Eirinaki

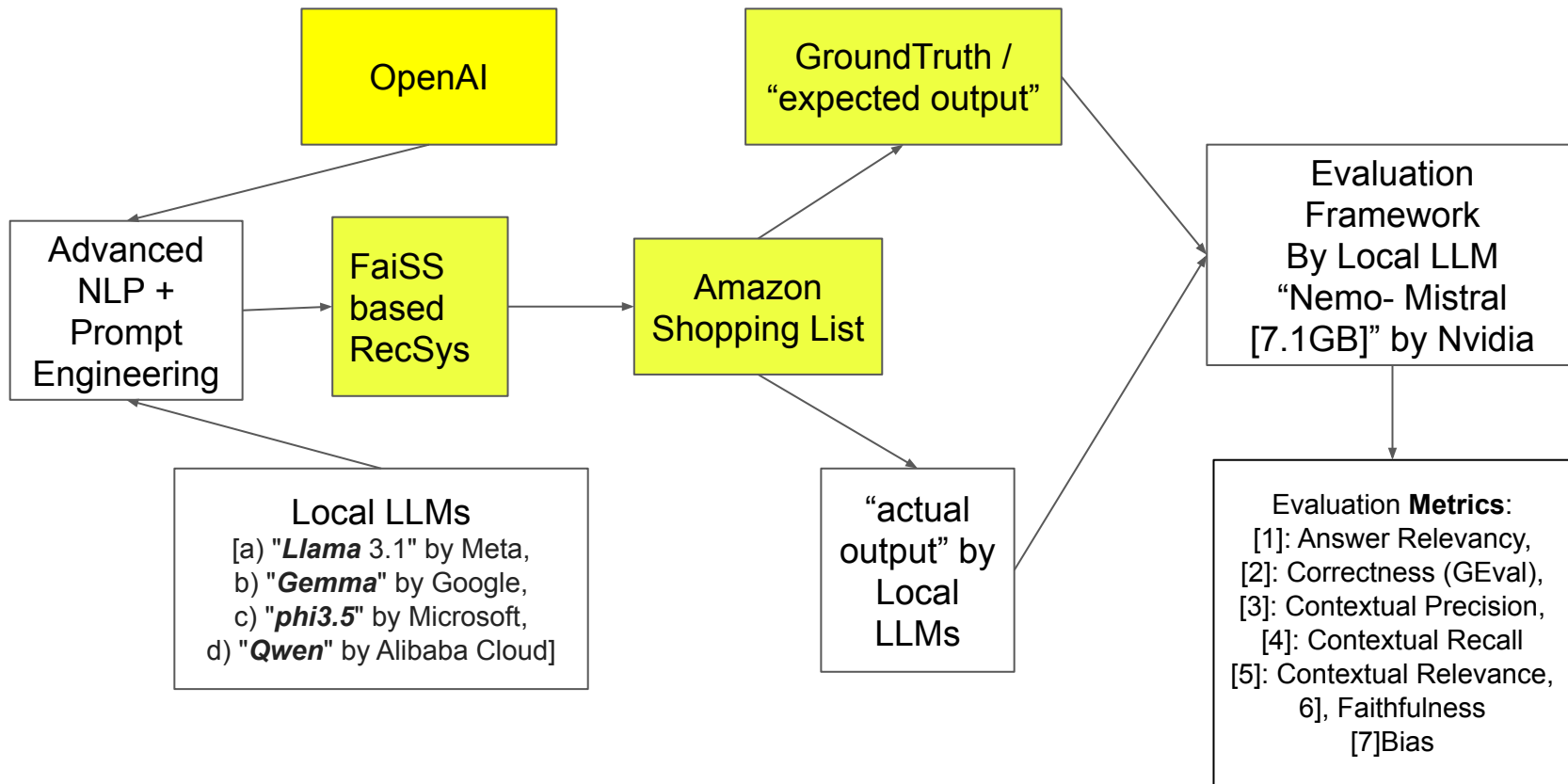
Perumal, Sasank
Vakkalagadda, Venkata Surya Kiran
Wanaskar, Kapil

San Jose State University

Introduction

- E-commerce Growth: The rapid expansion of e-commerce has greatly improved shopping convenience.
- Limitation in Personalization: Traditional customer support systems use predefined responses, often failing to meet individual customer needs.
- Integration of Generative AI: Incorporating Generative AI like OpenAI's GPT-4 and Meta's LLaMA with recommendation systems could revolutionize customer support by providing personalized and accurate responses.

Local LLM Evaluation: Experiment System Design



Metrics:

- Evaluated the Results using:
 1. Answer Relevancy
 2. Correctness (GEval)
 3. Contextual Precision
 4. Contextual Recall
 5. Contextual Relevancy
 6. Faithfulness
 7. Bias

Metrics



Introduction

G-Eval

Prompt Alignment

Answer Relevancy

Faithfulness

Contextual Precision

Contextual Recall

Contextual Relevancy

Json Correctness

Tool Correctness

Summarization

Hallucination

Bias

Toxicity

RAGAS

Custom Metrics

Conversational Metrics



Multimodal Metrics



Baseline and Implementation

- For Baseline Approach we have used human evaluated and GPT4 generated recommendations
- Implemented it on 4 different Local LLM Models
 - [a) "Llama 3.1" by Meta,
 - b) "Gemma" by Google,
 - c) "phi3.5" by Microsoft,
 - d) "Qwen" by Alibaba Cloud]
 - e) ~Wizardlm2~

LlaMa Implementation

```
74
75 # User input
76 input_dish = st.text_input("Enter the dish you want to cook:",
77                             placeholder="Enter a dish here...")
78
79 if st.button("Recommend Shopping List"):
80     # Get only 3 main ingredients
81     main_ingredients = get_main_ingredients(input_dish)
82
83     # Display main ingredients
84     st.write("1. Main Ingredients:")
85     for ingredient in main_ingredients:
86         st.write(f"• {ingredient}")
87
88     # Get and display recommendations for main ingredients
89     st.write("\n2. Items for Each Main Ingredient:")
90     for ingredient in main_ingredients:
91         recommendations = get_product_recommendations(ingredient, docsearch)
92         if recommendations:
93             st.write(f"• {ingredient}:")
94             for i, item in enumerate(recommendations, 1):
95                 st.write(f"    {i}. {item}")
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS 35

python3 - ground_truth_dataset

External URL: http://130.65.254.10:8533

2024-11-25 23:52:19.926 Session with id cbd738f6-2df2-4045-a975-72ddc37ed6ac is already connected! Connecting to a new session.

~2

[33]+ Stopped streamlit run test_llama_3.1.py

o (llama_env) 016701321@coe-csps-206-18:~/kapil_files/ground_truth_dataset\$ streamlit run test_llama_3.1.py

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8534

Network URL: http://10.31.96.168:8534

External URL: http://130.65.254.10:8534

With Llama 3.1 8B

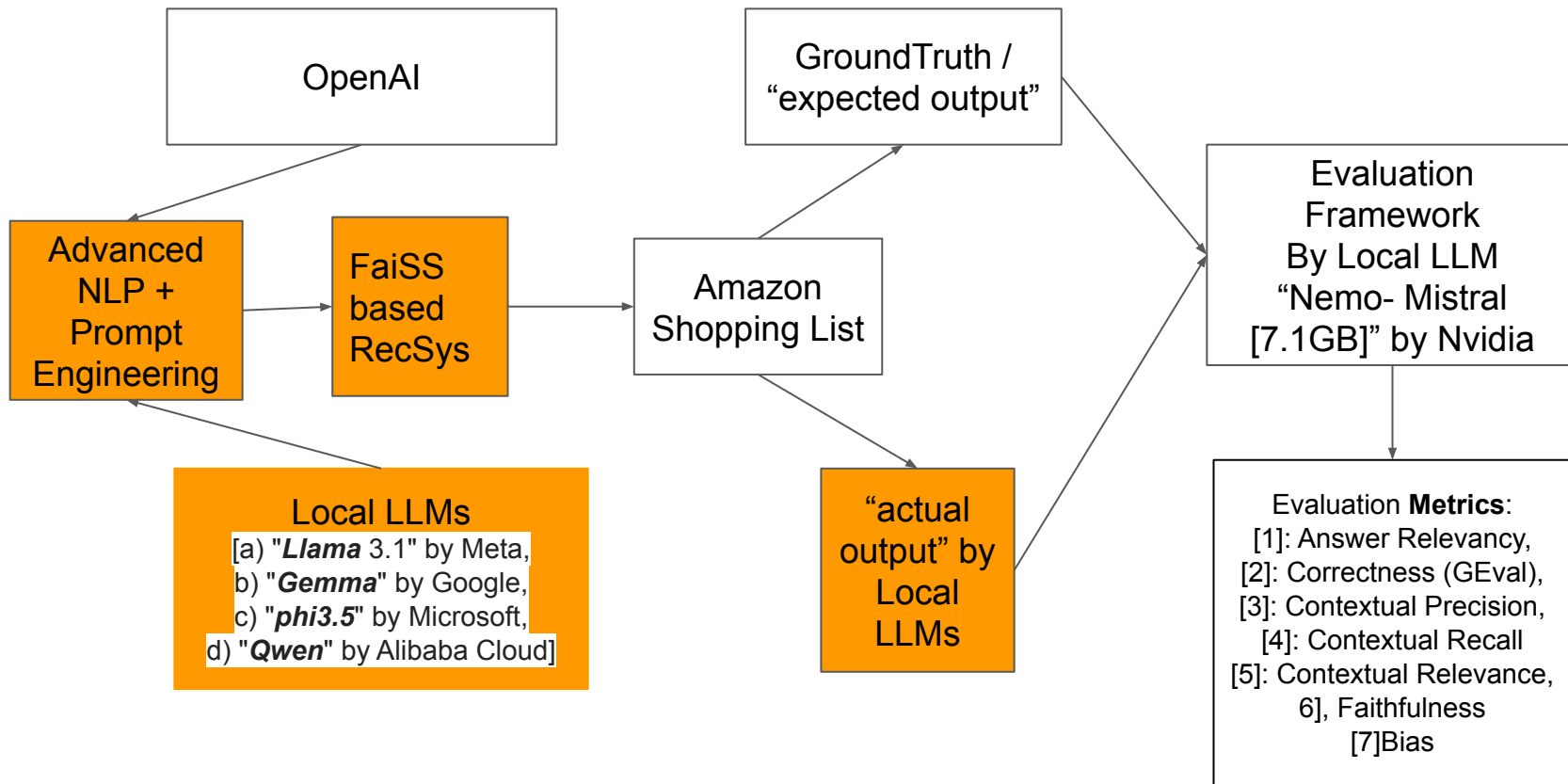
Enter the dish you want to cook:

cookie

Recommend Shopping List

1. Main Ingredients:
 - Flour
 - Butter
 - Sugar
2. Items for Each Main Ingredient:
 - Flour:
 1. title: Cocoyam Fu Fu Flour
 2. title: Golden Semolina And Extra Fancy Durum Pasta Flour
 - Butter:
 1. title: Historic Lychburg Tennessee Whiskey Sweet Tater Butter
 2. title: Apple Butter
 - Sugar:
 1. title: CX Products 4 Ounce Black Crystal Sugar
 2. title: Dynasty Sugar Ginger

Local LLM Evaluation: Experiment System Design



```
kapil_files
> 295A
v ground_truth_dataset
> 1_env
> faiss_index
> faiss_index_gemma_2b
> faiss_index_llama
> faiss_index_llamalates
> faiss_index_openai
> faiss_index2
> llama_env
> Llama-3.1-8B
0_app_5_streamlit.py
1_local_embeddings.py
2_local_embeddings.py
3_local_embeddings.py
4_surya_openAI_ingredients_faiss_se...
data_processing.ipynb
gemma_2b.py
gemma2_9b.py
{} Grocery_and_Gourmet_Food_5.json
[] Grocery_and_Gourmet_Food_filtered_...
llama_3.1.py
[] raw_titles.csv
[] requirements.txt
> LLM_Evaluation
> ollama_LLM_models
> venv_1
① README.md
LLM_recommendation_Amazon_fo...
.bash_history
.bash_logout
.bashrc
.gitconfig
.profile
...

20 recommendations(ingredient, docsearch):
28 :
29 , 'page_content'):
31 ot in seen:
36
37 mmendations
38
39 app
40 ing Recommendation with Gemma2: 9b")
41
42
43 iss_index"
44 itles.csv"
45
46
47 ss_index():
48 index if it exists, otherwise create and save i
49 FAISS_INDEX_PATH):
50 FAISS.load_local(
51 X_PATH,
52 ddings(model="gemma2"), # Removed batch_size h
53 erous_deserialization=True
54
55
56 ader(
57 CSV_FILE_PATH,
58
59
...
python3 - ground_truth_dataset + v [] [] ... ^ x
(llama_env) 016701321@coe-cmpe-206-18:~/kapil_files/ground_tr
uth_dataset$ streamlit run gemma2_9b.py

Collecting usage statistics. To deactivate, set browser.gathe
rUsageStats to false.

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8509
Network URL: http://10.31.96.168:8509
External URL: http://130.65.254.10:8509

[]
```

Amazon Shopping Recommendation with Gemma2: 9b

Enter the dish you want to cook:

lemon cake

Recommend Shopping List

1. Main Ingredients:

- Flour
- sugar
- lemons

2. Items for Each Main Ingredient:

- Flour:
 1. title: Authentic Foods Garfava Flour
 2. title: White Whole Wheat Flour
- sugar:
 1. title: Bigelow Oolong Tea Bags Classic- 20 ct Caffeinated Individual Black Tea Bags, for Hot Tea or Iced Tea, Drink Plain or Sweetened with Honey or Sugar
 2. title: Food to Live Macadamia Nuts (Raw, Kosher) (8 Ounces)
- lemons:

Gemma2 Implementation


```
ground_truth_dataset
> 1_env
> faiss_index
> faiss_index_gemma_2b
> faiss_index_gemma2_9b
> faiss_index_llama
> faiss_index_llamalates
> faiss_index_openai
> faiss_index2
> llama_env
> Llama-3.1-8B
0_app_5_streamlit.py
1_local_embeddings.py
2_local_embeddings.py
3_local_embeddings.py
4_surya_openAI_ingredients_faiss_se...
data_processing.ipynb
gemma_2b.py
gemma2_9b.py
{} Grocery_and_Gourmet_Food_5.json
Grocery_and_Gourmet_Food_filtered...
llama_3.1.py
llama_32.py
raw_titles.csv
requirements.txt
> LLM_Evaluation
> ollama_LLM_models
> venv_1
① README.md
LLM_recommendation_Amazon_fo...
.bash_history
.bash_logout
.bashrc
.gitconfig
.profile

79 st.button("Recommend Shopping List"):
80
81 # Display main ingredients
82 st.write("1. Main Ingredients:")
83 for ingredient in main_ingredients:
84     st.write(f"* {ingredient}")
85
86 # Get and display recommendations for main ingredier
87 st.write("\n2. Items for Each Main Ingredient:")
88 for ingredient in main_ingredients:
89     recommendations = get_product_recommendations(ingredient)
90     if recommendations:
91         st.write(f"* {ingredient}:")
92         for i, item in enumerate(recommendations, 1):
93             st.write(f"    {i}. {item}")

TERMINAL ... python3 - ground_truth_dataset + - □ □ ... ^ x
Try 'streamlit run --help' for help.

Error: Invalid value: File does not exist: llama_3.2.py
(ollama_env) 016701321@coe-cmpe-206-18:~/kapil_files/ground_truth_data$ streamlit run llama_32.py
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8510
Network URL: http://10.31.96.168:8510
External URL: http://130.65.254.10:8510
```

Amazon Shopping Recommendation with Llama 3.2

Enter the dish you want to cook:

Tamarind Rice

Recommend Shopping List

1. Main Ingredients:

- Tamarind paste
- long grain rice
- ghee

2. Items for Each Main Ingredient:

• Tamarind paste:

1. title: Walkerswood Spicy Jamaican Solomon Gundy Smoked Herring Paste

2. title: Garlic Paste

• long grain rice:

1. title: 2.2 Lb Molinella Arborio Rice

2. title: Vigo Saffron Yellow Rice

• ghee:

1. title: Yogurt Seasoning an Indian raita spice, 1 jar

2. title: Assorted Rock Candy strings

LLaMa 3.2 Implementation

```
FOLDERS: 016701321@... tere20000.csv 4_surya_openAI_ingredients_faiss_searchinDB.py llama_3.1.py llama_3.2.py mistral_nemo.py x ...

> .cache
> .keras
> .nv
> .ollama
> .vscode-server
v kapil_files
> 295A
v ground_truth_dataset
> 1_env
> faiss_index_gemima_2b
> faiss_index_gemima2_9b
> faiss_index_llama
> faiss_index_llama32
> faiss_index_llamalates
> faiss_index_mistral_nemo
> faiss_index_openai
> faiss_index2
> hugface
> llama_env
> Llama-3.1-8B
> 0_app_5_streamlit.py
> 1_local_embeddings.py
> 2_local_embeddings.py
> 3_local_embeddings.py
> 4_surya_openAI_ingredients_faiss...
data_processing.ipynb
> gemma_2b.py
> gemma2_9b.py
(i) Grocery_and_Gourmet_Food_5.json
Grocery_and_Gourmet_Food_fite...
> llama_3.1.py
> llama_3.2.py
> mistral_nemo.py
> raw_titles.csv
> requirements.txt
> LLM_Evaluation
> ollama_LLM_models

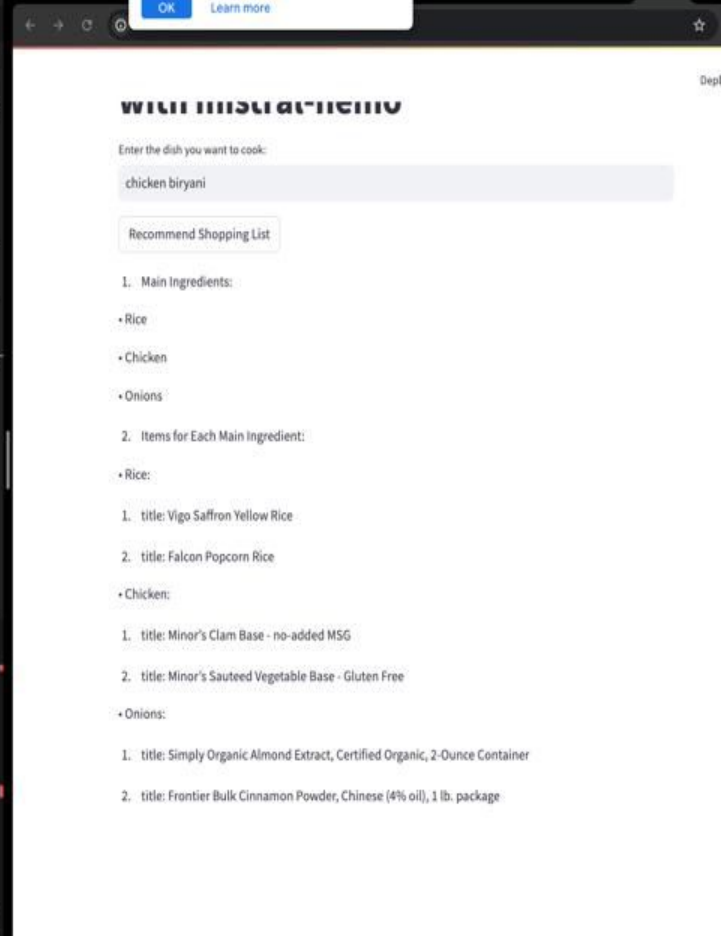
kapil_files > ground_truth_dataset > mistral_nemo.py
47 def load_or_create_faiss_index():
48     """Load the FAISS index if it exists, otherwise create and save it."""
49     if os.path.exists(FAISS_INDEX_PATH):
50         faiss_index = FAISS.load_local(
51             FAISS_INDEX_PATH,
52             OllamaEmbeddings(model="mistral-nemo"), # Removed batch_size here
53             allow_dangerous_deserialization=True
54         )
55     else:
56         loader = CSVLoader(
57             file_path=CSV_FILE_PATH,
58             csv_args={
59                 'delimiter': ',',
60                 'quotechar': '"',
61                 'fieldnames': ['title']
62             }
63         )
64         docs = loader.load()
65         embeddings_model = OllamaEmbeddings(model="mistral-nemo") # Removed batch_size here
66         faiss_index = FAISS.from_documents(docs, embeddings_model)
67         faiss_index.save_local(FAISS_INDEX_PATH) # Save the new index
68     return faiss_index
69
70
71 # Load the document search
72 with st.spinner('Loading data...'):
73     docsearch = load_or_create_faiss_index()
74
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 35
bash - ground_truth_dataset
^Z
[2]+ Stopped streamlit run mistral_nemo.py
((llama_env) 016701321@coe-cmpe-206-18:~/kapil_files/ground_truth_datasets$ streamlit run mistral_nemo.py

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8503
Network URL: http://10.31.96.168:8503
External URL: http://130.65.254.10:8503

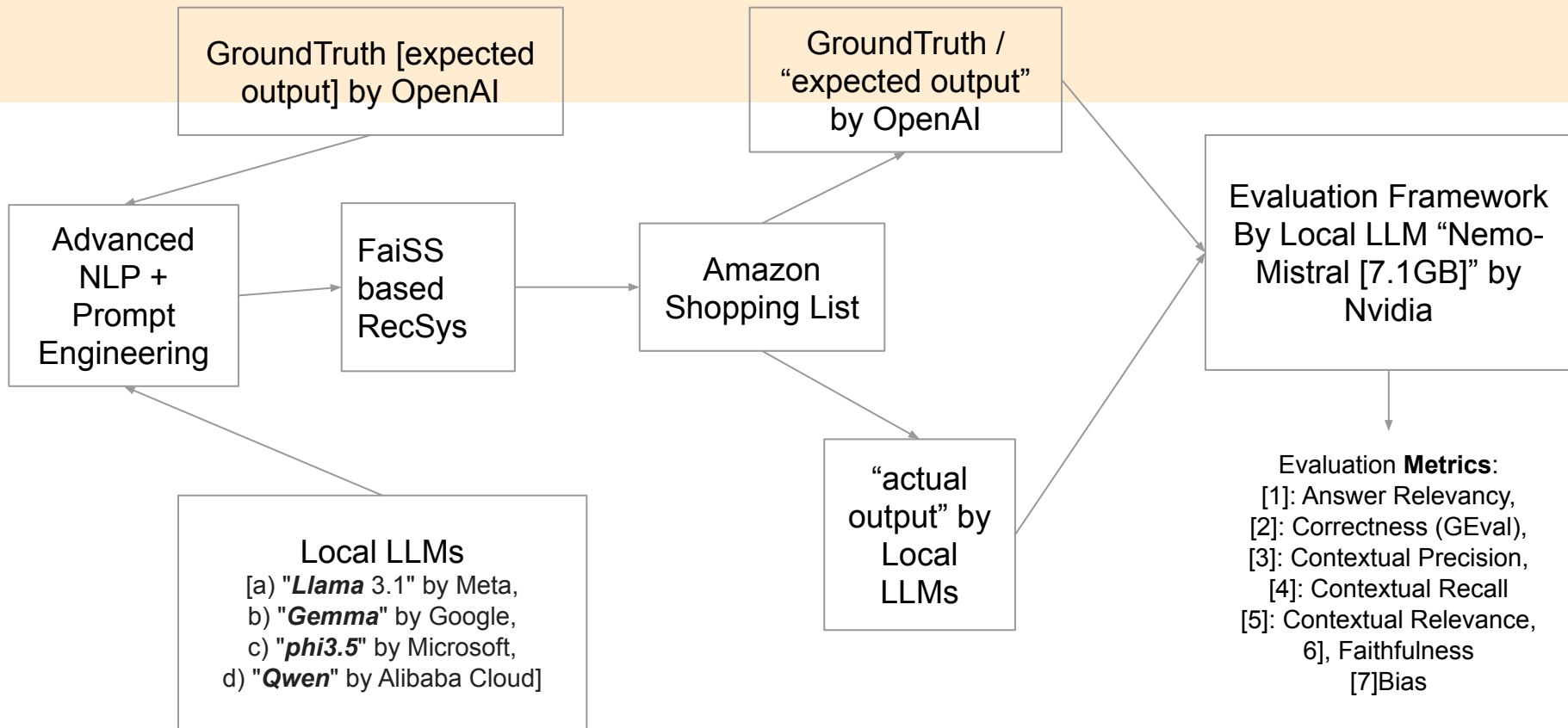
^Z
[3]+ Stopped streamlit run mistral_nemo.py
((llama_env) 016701321@coe-cmpe-206-18:~/kapil_files/ground_truth_datasets$
```



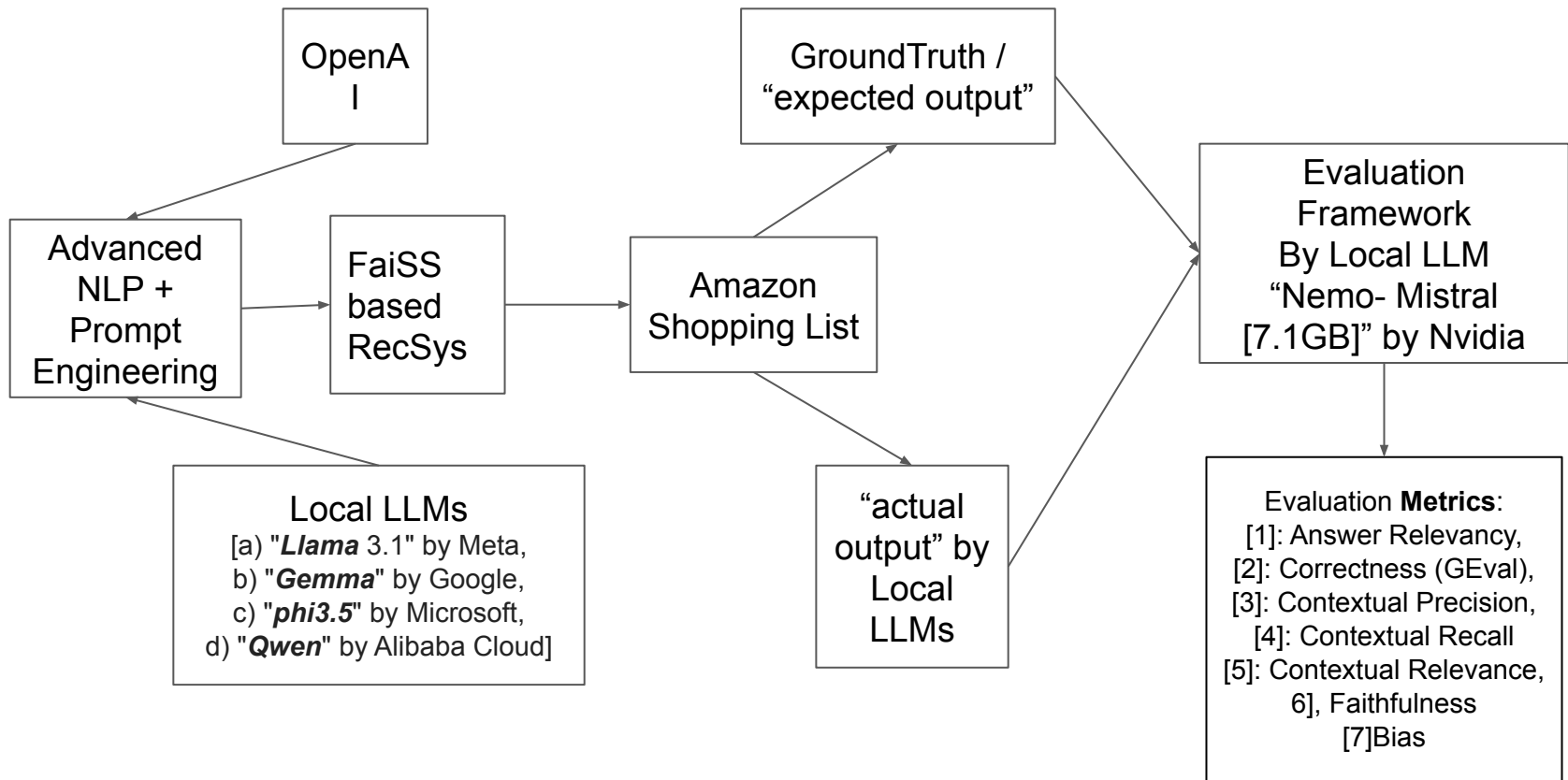
Mistral Nemo Implementation

Local LLM Evaluation: Experiment System Design

11



Local LLM Evaluation: Experiment System Design



	Contextual Recall	0unces, Fresh Italian Parsley, Fresh Dil Weed', error=None) 0.25 (threshold=0.5, evaluation model=gpt-4, reason=The score is 0.25 because the 1st node in the retrieval context only mentions 'olive oil' which matches with 10 OLIVE OIL & VINEGAR Organic Crushed Meyer Lemon Olive Oil, 8.45 Fluid Ounce' in the expected output. However, it does not contain any information related to 'Omaha Steaks or Mixed VegetablesFresh Spinach Leaves', 'Frontier Organic Cilantro Leaf - Flakes - 0.56 Ounces', or 'Fresh Italian Parsley, Fresh Dil Weed' that are also present in the expected output., error=None)	FAILED	
	Contextual Relevancy	1.0 (threshold=0.5, evaluation model=gpt-4, reason=The score is 1.00 because the retrieval context perfectly matches the input, as evidenced by the relevant statement: 'Salata Horta is a Greek salad made with wild greens, olive oil, and lemon'. There are no stated reasons for irrelevancy., error=None)	PASSED	
	Faithfulness	1.0 (threshold=0.5, evaluation model=gpt-4, reason=The score is 1.00 because there are no contradictions between the actual output and the retrieval context., error=None)	PASSED	
	Bias	0.0 (threshold=0.5, evaluation model=gpt-4, reason=The score is 0.00 because the actual output is perfectly balanced and free of bias, demonstrating exemplary fairness and objectivity., error=None)	PASSED	

Total estimated evaluation tokens cost: 1.431995 USD

Results saved in /kapil_wanaskar/LLM_Evaluation/DEEPEVAL_RESULTS_FOLDER_16_1 as 20241126_004635

✓ Tests finished 🎉! View results on

<https://app.confident-ai.com/project/cm2qy1gvr0e3ckyq11dly18sh/evaluation-n-testing/test-runs/cm3y7qeqx0021c4l7zmr1m0g2/test-cases>.

!! Friendly reminder 📢: You can also run evaluations with ALL of deepeval's metrics directly on Confident AI instead.

(2_venv deepeval) 016649880@coe-cmpe-206-18:/kapil_wanaskar/LLM_Evaluation\$



G07 Customer Support Enhancement in E-commerce with Generative Artificial Intelligence and Recommendation Systems: Evaluating Large Language Models Project Advisor: Dr. Magdalini Eirinaki

Wanaskar, Kapil (MS Artificial Intelligence)
Perumal, Sasank (MS Artificial Intelligence)
Vakkalagadda, Venkata Surya Kiran Naidu (MS
Artificial Intelligence)

Introduction

- E-commerce Growth:** The rapid expansion of e-commerce has greatly improved shopping convenience.
- Limitation in Personalization:** Traditional customer support systems use predefined responses, often failing to meet individual customer needs.
- Integration of Generative AI:** Incorporating Generative AI like OpenAI's GPT-4 and Meta's LLaMA with recommendation systems could revolutionize customer support by providing personalized and accurate responses.
- Advanced System Features:** These systems use natural language understanding from LLMs and similarity searches from recommendation modules to deliver tailored product suggestions, enhancing chatbot interactions.
- Limitations of Current Systems:** E-commerce support systems handle simple queries well but falter with complex issues and personalization. Hybrid systems also struggle to scale and maintain context.
- Recommendation System Challenges:** Traditional recommendation methods struggle with new users and offer limited variety, underscoring the need for a more dynamic solution to improve user experience.

Implementation:

- Combines Gen AI with advanced recommendation technologies and vector databases to address current limitations.
- Utilizes LLM's, such as LLaMa 3.1, Wizard LM-2, Qwen2, Gemma2, LLaMa 3.2.
- Utilizes a FAISS-based vector database for fast and relevant product recommendations based on user data.
- Features a user-friendly chatbot interface for seamless interaction with model output and queries.
- Conducts thorough testing using metrics like relevance, meets performance benchmark.

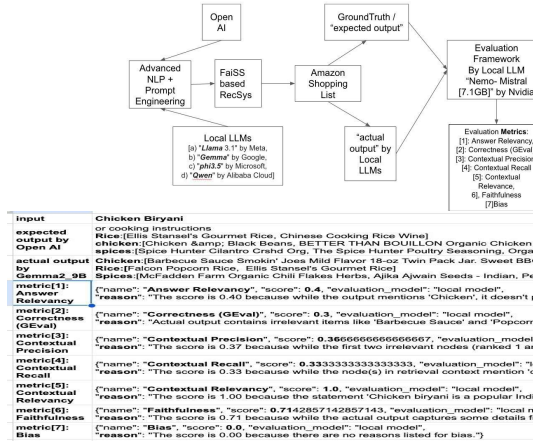
Dataset

Dataset Used : Amazon Grocery and Gourmet

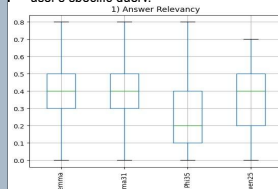
- Rich Review Data:** Includes detailed customer reviews, ratings, and timestamps for grocery and gourmet food products.
- Diverse Product Information:** Covers product attributes like name, price, brand, and categories.
- Applications:** Used for sentiment analysis, recommendation systems, and consumer behavior studies.

Project Architecture

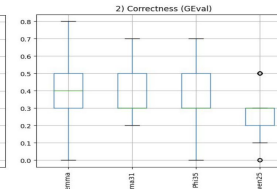
Local LLM Evaluation: Experiment System Design



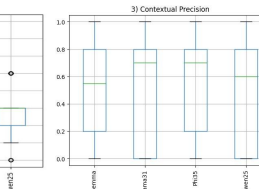
Answer Relevance: Measure how well the system's response match the user's specific query.



Correctness: measures the factual accuracy of the system's responses

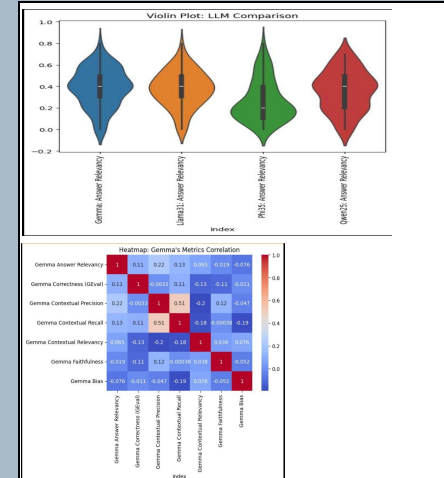


Contextual Precision: Assesses the accuracy of the information provided in the context of the user's query.



- Example output generated for the evaluation metrics by a Local LLM.
- The outputs generated by other LLMs are evaluated against the ground truth which is generated by GPT4 and Human Evaluated.
- Responses are evaluated based on these metrics:
 - Answer Relevance
 - Correctness
 - Contextual Precision
 - Contextual Recall
 - Contextual Relevance
 - Faithfulness

Analysis and Results



Metric Relationships: Highlights the correlation between evaluation metrics like contextual precision, recall, and correctness, with precision and recall showing the strongest positive correlation (0.51).

Metric Independence: Low correlations (near 0) between metrics like faithfulness, bias, and others indicate they capture distinct aspects of performance.

Visualization Insight: Summary/Conclusions vs. Independent The evaluation of Large Language Models in e-commerce highlights their potential to enhance efficiency and user satisfaction, with LLaMA 3.1 excelling in context sensitivity and metrics like Answer Relevance and Factual Correctness, though Qwen2.5 shows superior performance across most metrics.

Key References varies across models

- [1] A. Salemi and H. Zamani, "Evaluating retrieval quality in retrieval-augmented generation," arXiv.org, 4 2024.
- [2] J. B. Schafer, J. Konstan, and J. Riedl, "Recommender systems in e-commerce," pp. 168–166, 1999.

Acknowledgments

We express our deepest gratitude to Dr. Magdalini Eirinaki for her invaluable guidance and expertise throughout the development of the Customer Support Enhancement in E-commerce with Generative Artificial Intelligence and Recommendation Systems: Evaluating Large Language Models . Her expertise and insightful feedback have been crucial in refining our project and directing its progress.

Local LLM Models available for Evaluation

```
(venv_1) 016649880@coe-cmpe-206-18:/kapil_wanaskar$ ollama list
```

NAME	ID	SIZE	MODIFIED
phi3.5:latest	61819fb370a3	2.2 GB	2 days ago
mistral-nemo:latest	994f3b8b7801	7.1 GB	2 days ago
llama3.2:latest	a80c4f17acd5	2.0 GB	3 days ago
gemma2:latest	ff02c3702f32	5.4 GB	3 days ago
gemma:2b	b50d6c999e59	1.7 GB	3 days ago
gemma2:9b	ff02c3702f32	5.4 GB	3 days ago
llama3:8b	365c0bd3c000	4.7 GB	3 days ago
llama3.1:70b	c0df3564cfe8	39 GB	7 weeks ago
llama3.1:8b	42182419e950	4.7 GB	2 months ago
llama3.1:latest	42182419e950	4.7 GB	2 months ago

ValueError when tried weaker LLM model [Llama3.1] for Evaluation

```
rations.)  
est summary info =====  
ot[test_case0] - ValueError  
ot[test_case1] - ValueError  
ot[test_case2] - Failed: Met  
ot[test_case3] - ValueError  
ot[test_case4] - Failed: Met
```


Evaluation pipeline With Mistral-Nemo [7.1 GB] model

🎉 Congratulations! You're now using a local model for all evals that require an LLM.

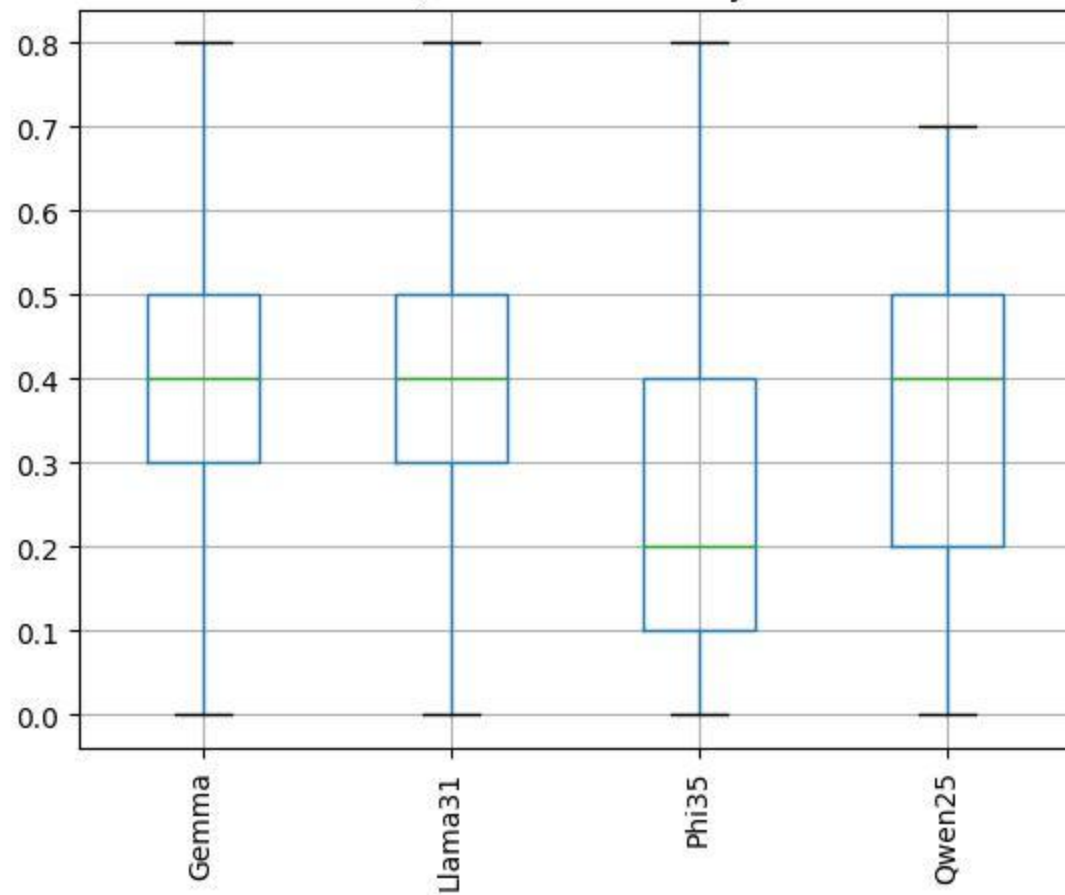
```
○ (2_venv_deepaval) 016649880@coe-cmpe-206-18:/kapil_wanaskar/LLM_Evaluation$  
● (2_venv_deepaval) 016649880@coe-cmpe-206-18:/kapil_wanaskar/LLM_Evaluation$ export DEEPEVAL_RESULTS_FOLDER="/kapil_wanaskar/LLM_Evaluation/dataset/Llama31"  
○ (2_venv_deepaval) 016649880@coe-cmpe-206-18:/kapil_wanaskar/LLM_Evaluation$ deepeval  
test run test_19_llama31_to_MistralNemo.py  
Evaluating 1 test case(s) in parallel: |100% (1/1) [Time Taken: 01:36, 96.85s/test  
Evaluating 1 test case(s) in parallel: |100% (1/1) [Time Taken: 01:17, 77.64s/test case]  
Evaluating 1 test case(s) in parallel: |100% (1/1) [Time Taken: 01:23, 83.84s/test case]  
Evaluating 1 test case(s) in parallel: | 0% (0/1) [Time Taken: 00:24, ?test case/s]  
Retry 5 times till the test passes  
Retry 1 attempt  
Evaluating 1 test case(s) in parallel: | 0% (0/1) [Time Taken: 00:53, ?test case/s]  
Evaluating 1 test case(s) in parallel: | 0% (0/1) [Time Taken: 00:00, ?test case/s]
```

```
0
  name : "test_customer_chatbot[test_case0]"
  input : "chicken biryani"
  actualOutput : "Recommended Shopping List:
  expectedOutput : " Royal Basmati Rice, 15-Po
retrievalContext
  success : false
metricsData
  0
  1
  2
  3
  4
  5
  6
  runDuration : 26.05272420808620
```

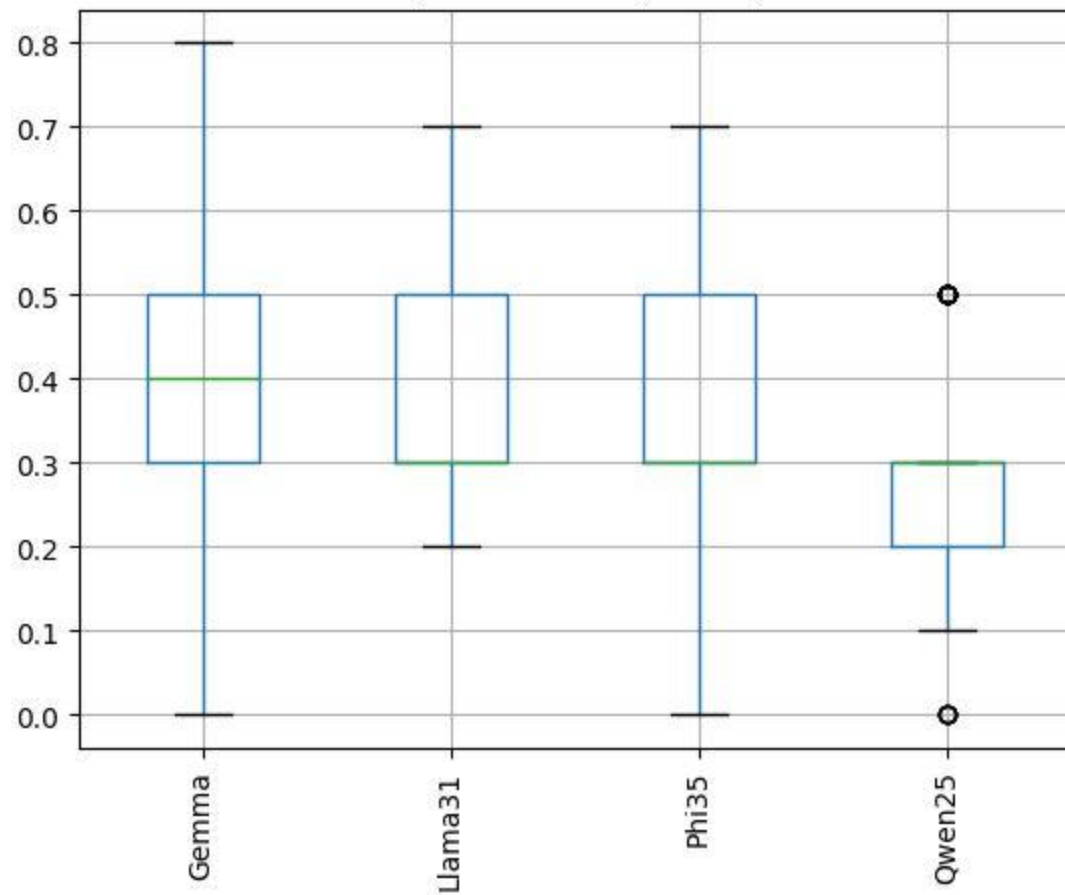
```
metricsData
  0
    name : "Answer Relevancy"
    threshold : 0.5
    success : true
    score : 1
    reason : "The score is 1.00 because the output pe
    strictMode : false
    evaluationModel : "gpt-4"
    evaluationCost : 0.06837
    verboseLogs : "Statements: [ "Recommended Sho
  1
    name : "Correctness (GEval)"
    threshold : 0.5
    success : false
    score : 0.028435262024017572
    reason : "The actual output does not align with the
    strictMode : false
    evaluationModel : "gpt-4"
    evaluationCost : 0.02541
    verboseLogs : "Criteria: None Evaluation Steps: [
  2
    name : "Contextual Precision"
    threshold : 0.5
    success : false
    score : 0
    reason : "The score is 0.00 because the first node
    strictMode : false
    evaluationModel : "gpt-4"
    evaluationCost : 0.03579
    verboseLogs : "Verdicts: [ { "verdict": "no", "reason
  3
    name : "Contextual Recall"
    threshold : 0.5
    success : false
    score : 0
    reason : "The score is 0.00 because none of the s
    strictMode : false
    evaluationModel : "gpt-4"
    evaluationCost : 0.04875
    verboseLogs : "Verdicts: [ { "verdict": "no", "reason
```

```
4
  name : "Contextual Relevancy"
  threshold : 0.5
  success : true
  score : 1
  reason : "The score is 1.00 because the retrieval context
  strictMode : false
  evaluationModel : "gpt-4"
  evaluationCost : 0.023189999999999995
  verboseLogs : "Verdicts: [ { "verdicts": [ { "statement": "C
5
  name : "Faithfulness"
  threshold : 0.5
  success : false
  score : 0.4
  reason : "The score is 0.40 because the actual output in
  strictMode : false
  evaluationModel : "gpt-4"
  evaluationCost : 0.09999
  verboseLogs : "Truths (limit=None): [ "Chicken Biryani is
6
  name : "Bias"
  threshold : 0.5
  success : true
  score : 0
  reason : "The score is 0.00 because the actual output is
  strictMode : false
  evaluationModel : "gpt-4o"
  evaluationCost : 0.00625
  verboseLogs : "Opinions: [ "While it's a tomato sauce, y
```

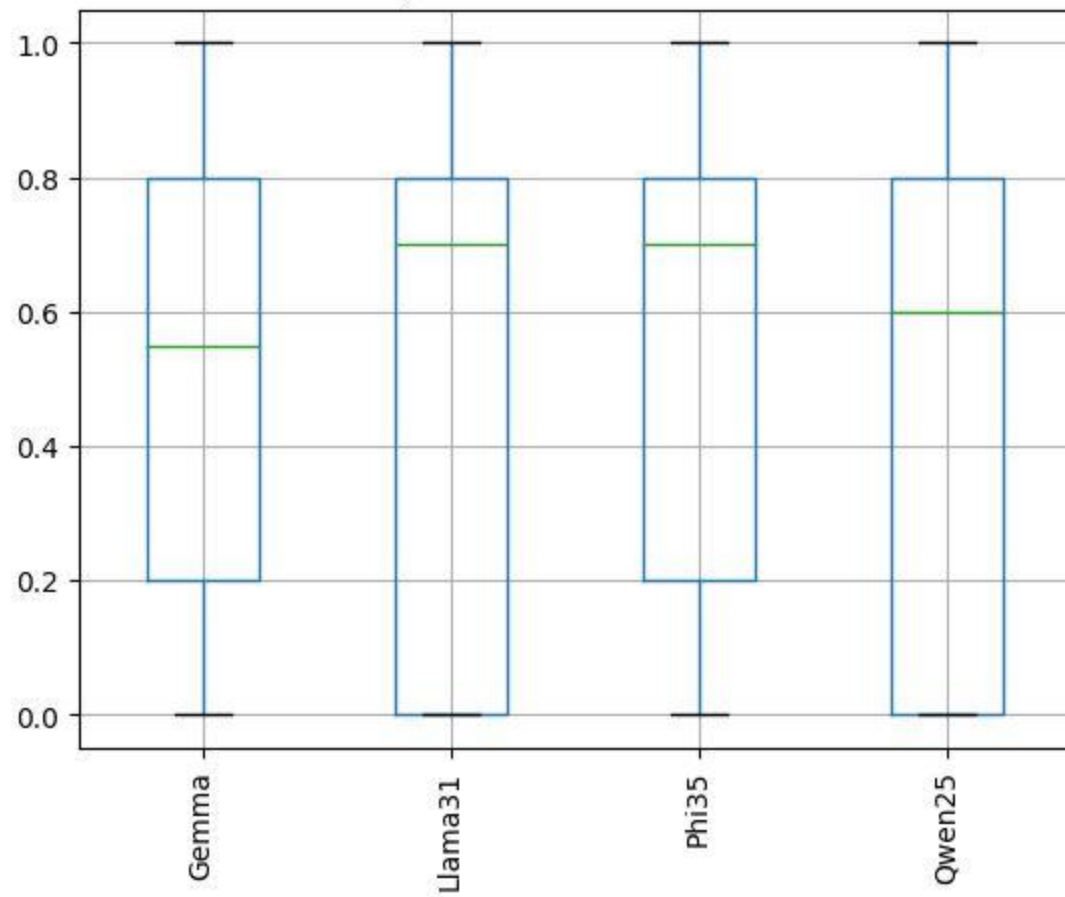
1) Answer Relevancy



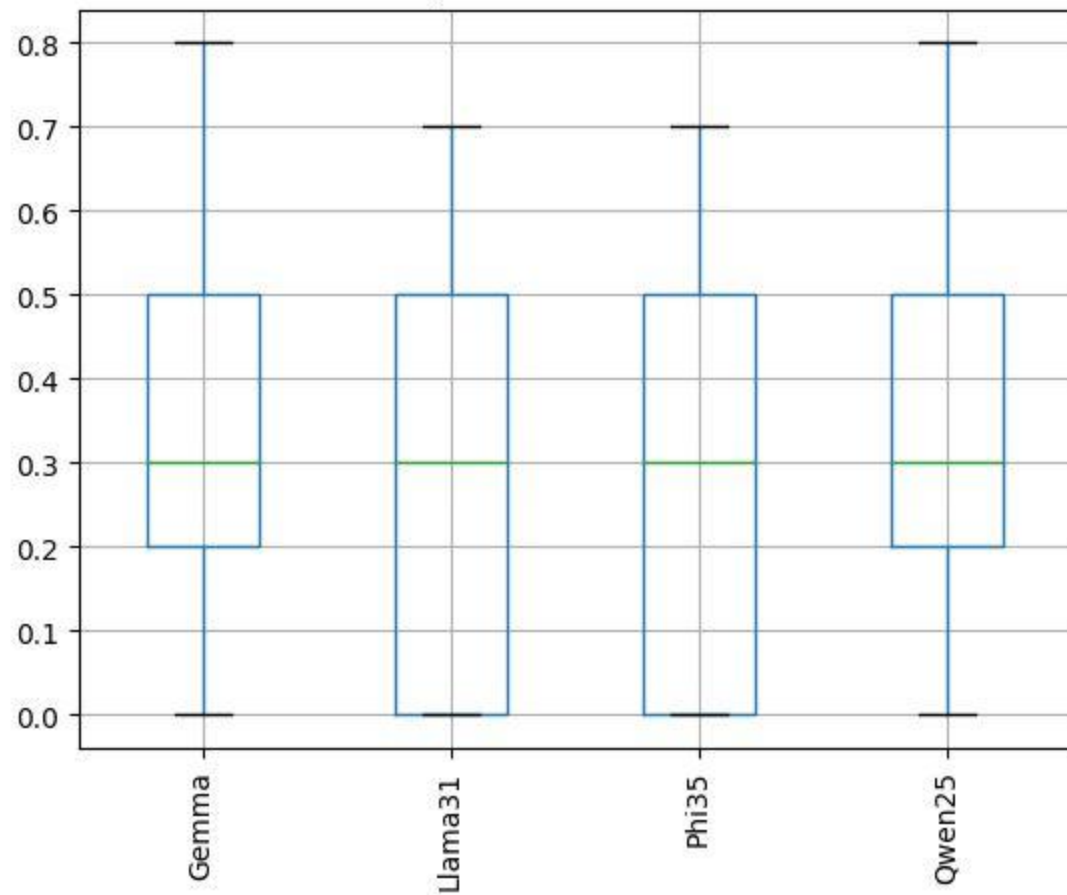
2) Correctness (GEval)



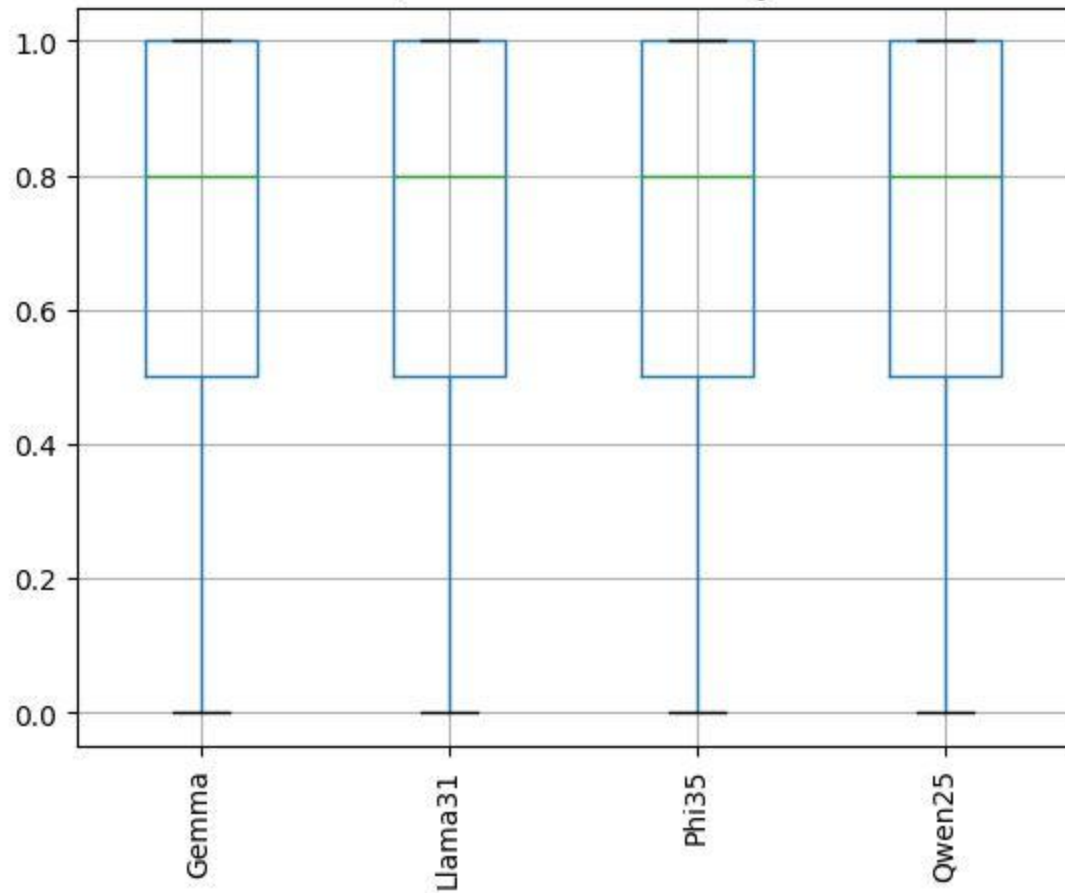
3) Contextual Precision



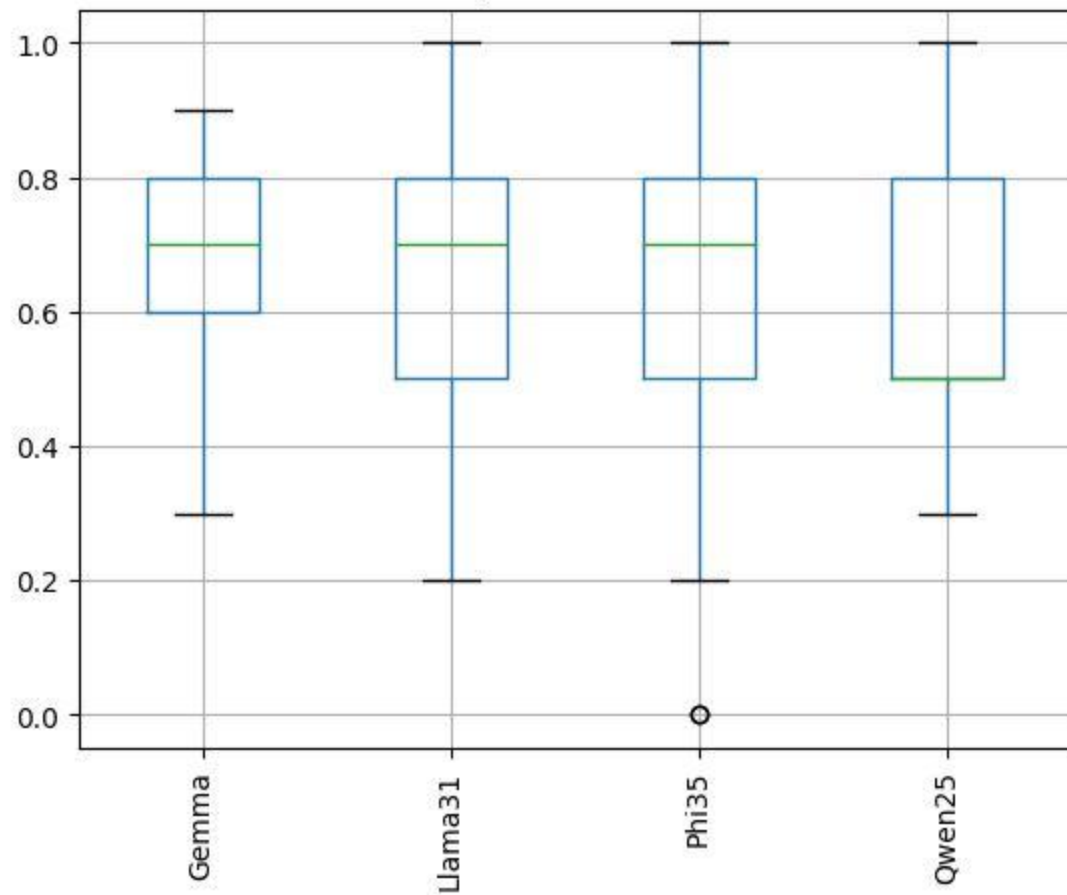
4) Contextual Recall



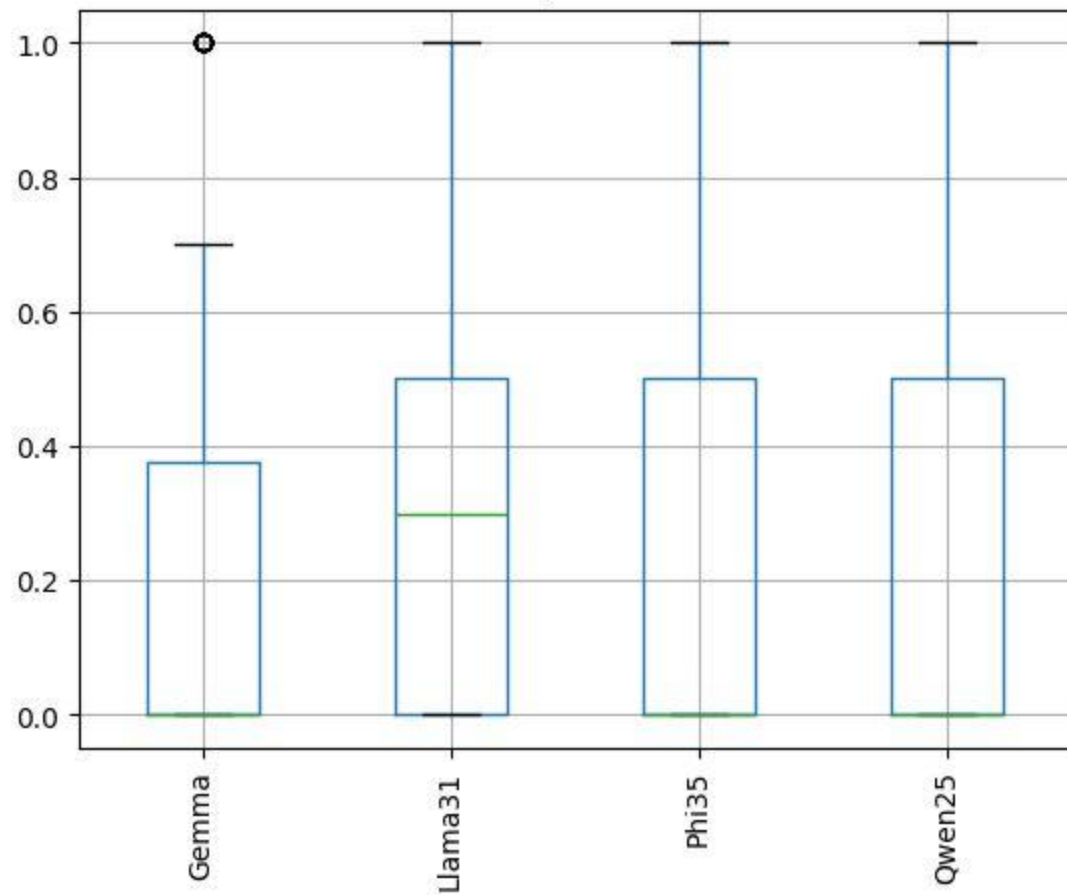
5) Contextual Relevancy



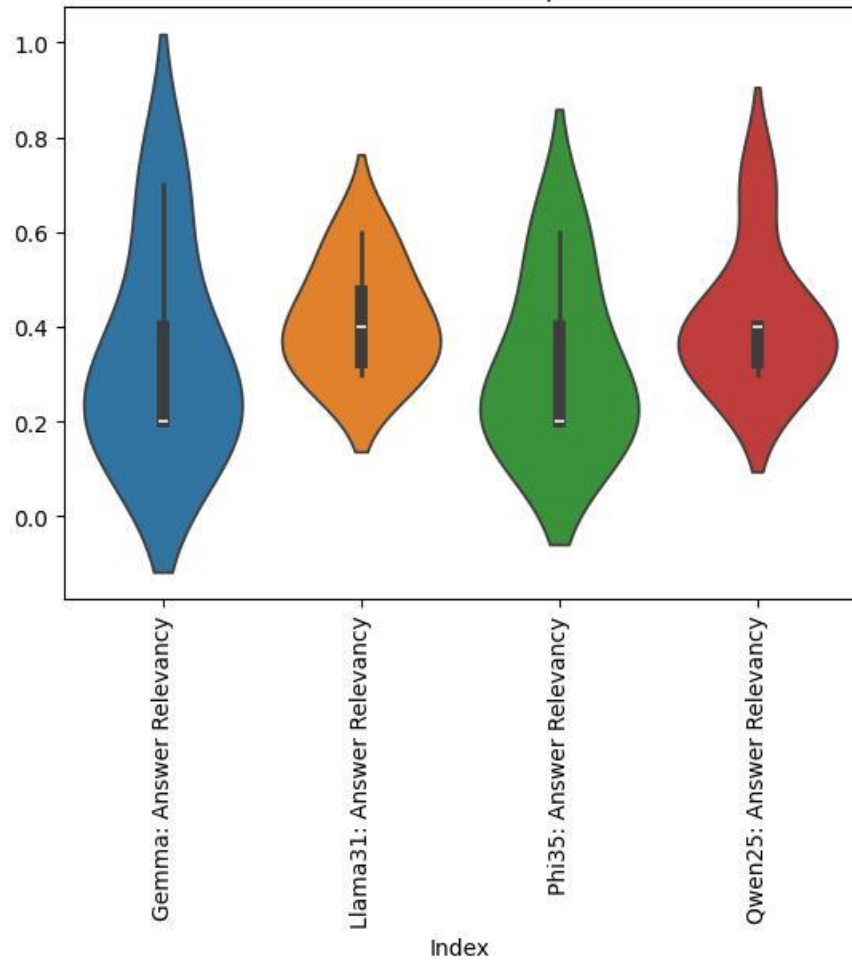
6) Faithfulness



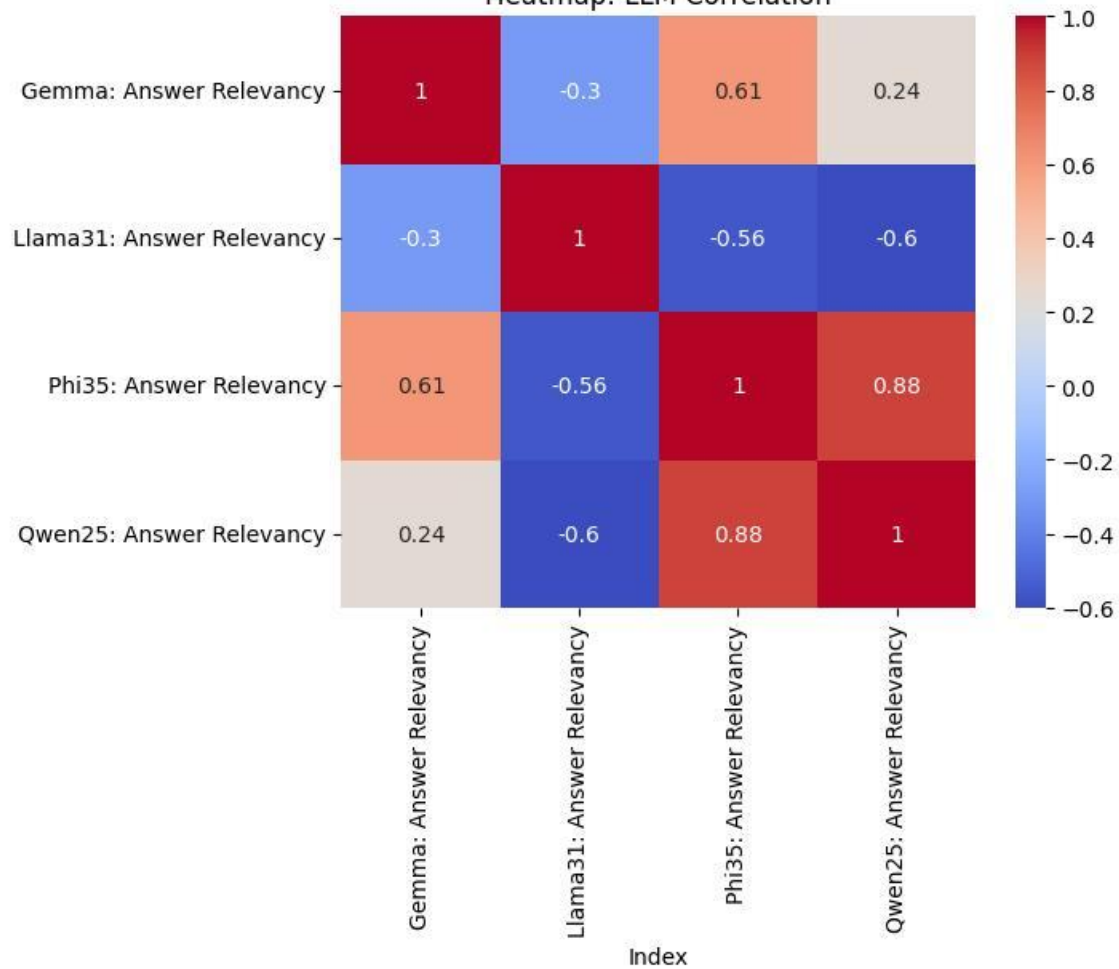
7) Bias



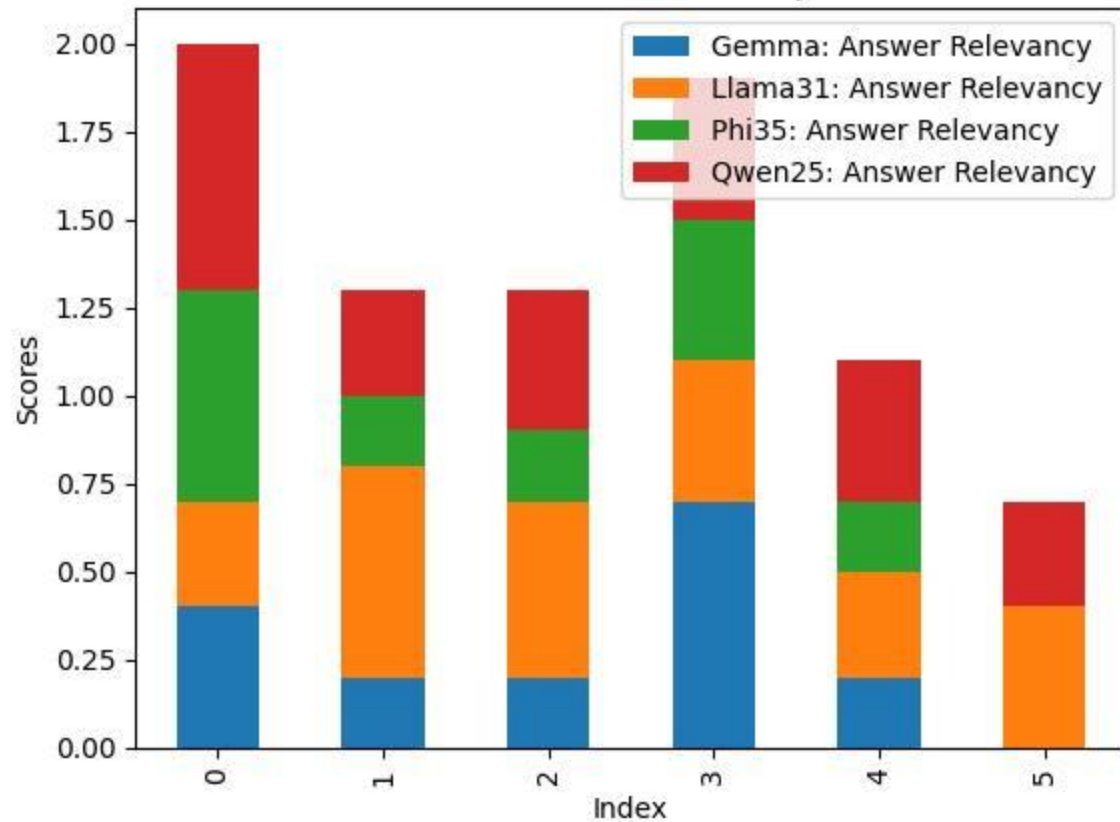
Violin Plot: LLM Comparison



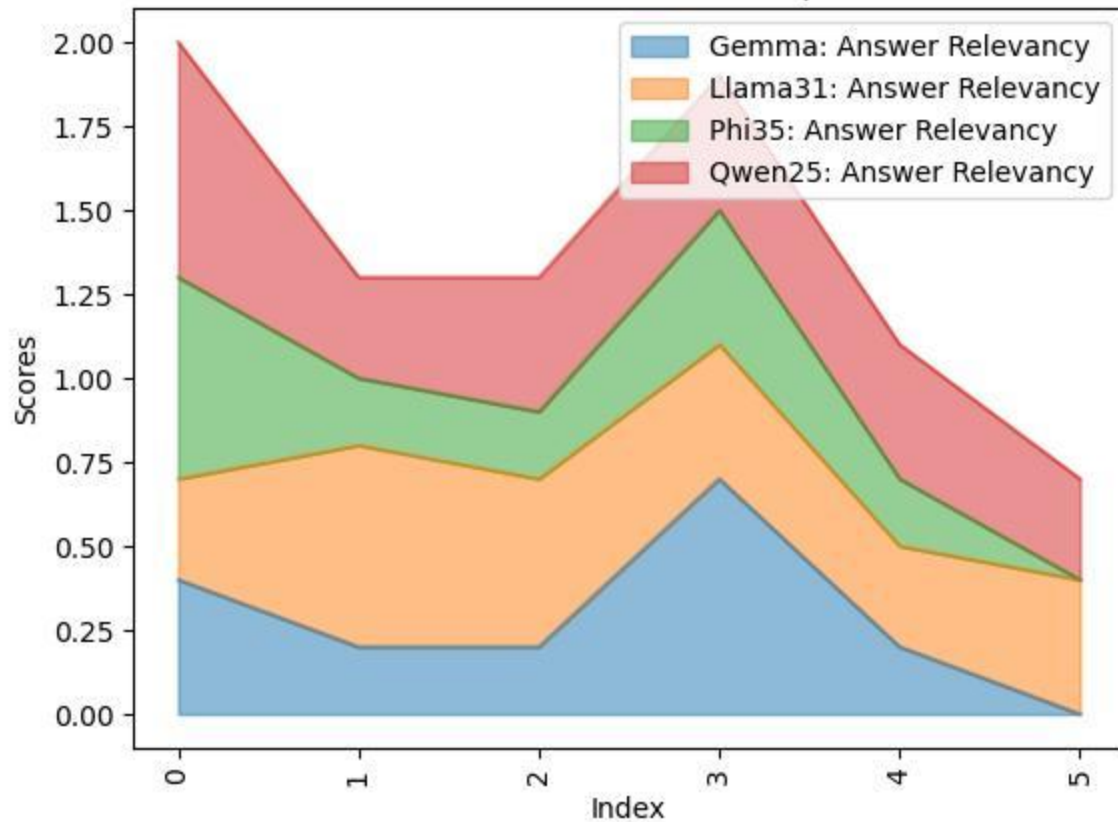
Heatmap: LLM Correlation



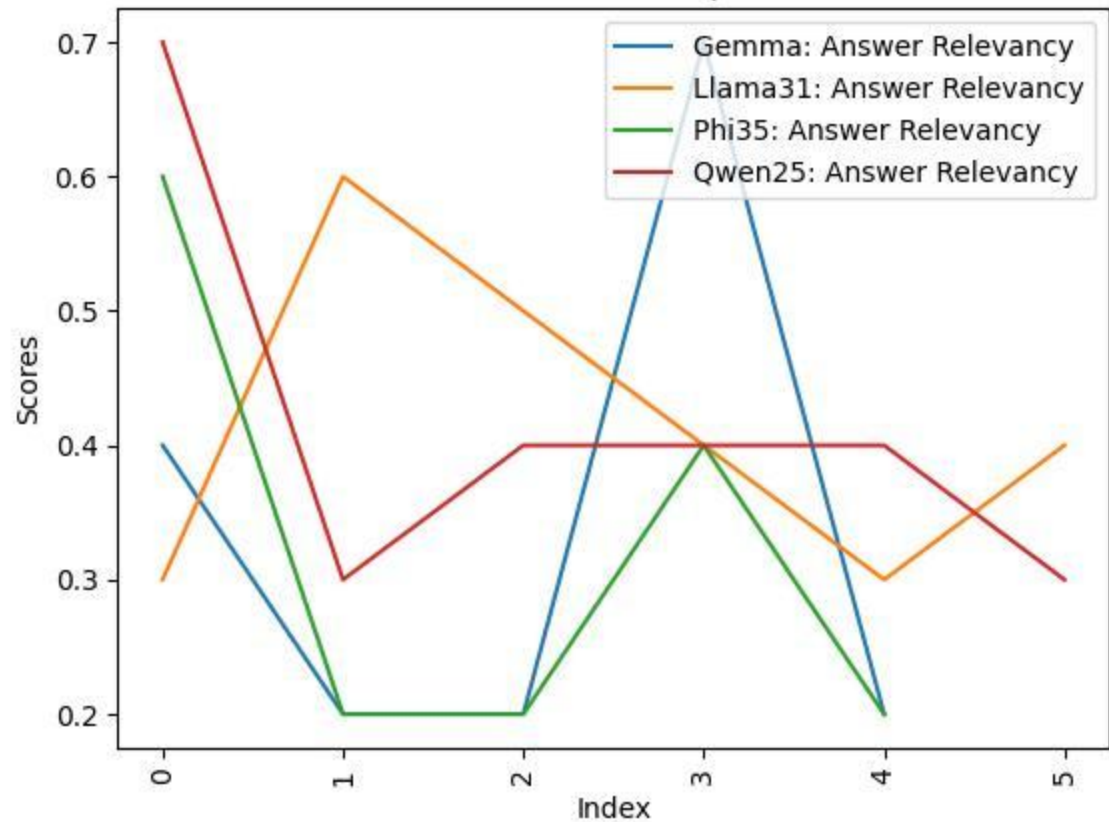
Stacked Bar Plot: LLM Comparison



Stacked Area Plot: LLM Comparison



Line Plot: LLM Comparison



DEMO

